

# Patient Profile

**Name:** Yash M. Patel

**Age:** 21

**Gender:** Not specified

## Summary of Results

The liver function panel (AST, ALT, ALP, Total Bilirubin, Direct Bilirubin) has not returned measurable values in the current report. Without numeric results, definitive conclusions cannot be drawn; however, the clinical context provided suggests a potential evaluation for non-alcoholic fatty liver disease (NAFLD) within the spectrum of metabolic liver disorders.

## Detailed Test Explanations

### AST (Aspartate Aminotransferase)

A pyridoxal-phosphate dependent enzyme found in hepatocytes, cardiac, and skeletal muscle. Elevated serum AST indicates hepatocellular injury; a disproportionately high AST/ALT ratio ( $>1$ ) may raise concern for alcoholic hepatitis, hepatocellular necrosis, or advanced liver disease. Result unavailable.

### ALT (Alanine Aminotransferase)

A primarily hepatic enzyme; elevations reflect hepatocellular damage. ALT typically exceeds AST in viral hepatitis, NAFLD, or medication-induced injury. Result unavailable.

### ALP (Alkaline Phosphatase)

An enzyme present in bile ducts, bone, and placenta. Elevated ALP suggests cholestasis or biliary obstruction; age-adjusted ranges differ in adolescents. Result unavailable.

### Total Bilirubin

Conjugated and unconjugated bilirubin reflect liver excretion and hemolysis. Elevated total bilirubin indicates impaired conjugation or excretion (e.g., cholestasis, hepatic impairment). Result unavailable.

### Direct Bilirubin

Measures conjugated bilirubin; elevations are more specific for cholestatic or excretory dysfunction. Result unavailable.

## Overall Interpretation

While numeric data are missing, the remarks emphasize NAFLD as a frequent cause of elevated AST/ALT in asymptomatic young adults. Given the absence of abnormal values, there is currently no biochemical evidence of hepatic injury. However, the patient may still be at risk if undiagnosed metabolic derangements exist.

## Recommendations

- **Repeat Liver Function Testing:** Perform a repeat LFT panel within 3–6 months or sooner if symptomatic (e.g., jaundice, fatigue).
- **Metabolic Screening:** Evaluate fasting glucose, HbA1c, lipid profile, and hepatic imaging (ultrasound) to assess for NAFLD.
- **Lifestyle Modifications:**
  - Adopt a balanced diet low in saturated fats and refined sugars.

- Aim for 150 minutes of moderate aerobic activity weekly.
- Maintain a healthy weight; a 5–10% reduction can improve liver enzymes.
- Limit alcohol intake to no more than 1 drink per day.
- **Follow-up:** Annual check-up with primary care provider and review of metabolic risk factors.

## Disclaimer

This report is informational and not a substitute for professional medical advice, diagnosis, or treatment. Always consult a qualified healthcare provider for personalized care.